

Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical AI

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Abstract—Hospital federated learning (FL) must simultaneously resist gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under multi-year retention. ZKFL-PQ composes NIST ML-KEM-768 transport; a dual-norm gate (ℓ_2 Unruh NIZK in $\mathcal{L}_\tau^{\text{bind}}$ plus public ℓ_∞ clipping on the same quantized vector); *Unruh-lifted* Enc-consistency for BFV coins; full-vector BFV opened by (t, n) -threshold partial decryption with a PartialDecrypt NIZK (malicious μ_i abort); round-transcript binding and adaptive public τ ; and default post-ZKP median. The artifact ships fused SEAL+threshold HE, shared-SIS Unruh, Keccak/SHA3 formal checks, and an innovation-pack evaluation. Empirically we report multi-seed baselines, UCI Breast Cancer under the innovation stack, MedM-NIST/ConvNet28, backdoor ASR, and a sparse-poison dual-norm ablation (100% reject vs. 0% for ℓ_2 -only). Complementary to Beskar-class secure aggregation.

Index Terms—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness

I. INTRODUCTION

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4]. Cryptographic accept alone is also insufficient: updates whose ℓ_2 norm respects a public threshold τ can still flip signs or implant triggers, so statistical robust aggregation must follow a successful proof.

a) Design goals.: We optimize for hospital consortia where (G1) ciphertext retention is measured in years, (G2) no single IT operator may hold a usable sk , (G3) oversized gradients must be rejectable without trusting the client, and (G4) in-bound Byzantine behavior that passes τ is handled by a robust aggregator rather than by tightening τ until honest

sites fail. These goals motivate the four-layer stack rather than any single primitive.

b) Contributions.:

- 1) A layered protocol that binds Unruh norm proofs to BFV ciphertexts via $\mathcal{L}_\tau^{\text{bind}}$ + *Unruh* Enc-consistency, opens with threshold partial decrypt *with* PartialDecrypt NIZK, then robust-aggregates (default median), with each layer addressing a distinct threat (Table I).
- 2) Dual-norm gating (ℓ_2 Unruh + public ℓ_∞ clip), adaptive public τ , and round-transcript binding—closing QROM-uniformity, sparse-poison, decryptor-malice, and cross-round swap gaps in prior residuals.
- 3) Concrete parameters, a named soundness/privacy theorem, Lean/Python/QROM.ec checks with SHA3/Keccak lane lemmas, and measured payloads under varying r and HE degree.
- 4) A reproducible artifact [12] and empirical study: multi-seed baselines, innovation-pack UCI run, MedM-NIST/ConvNet28, backdoor ASR, and dual-norm sparse-poison ablation.

II. THREAT MODEL AND CLAIMS

Adversary: (i) a network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f < N/2$ Byzantine clients that may submit arbitrary vectors and proofs; (iii) a curious aggregator that sees all protocol messages but does not hold sk alone; (iv) up to $t-1$ colluding threshold decryptors, *plus* the possibility that one decryptor returns a malformed partial μ_i .

Trust. Honest clients follow local SGD, ℓ_∞ clipping, and the Prove interface. Honest decryptors compute μ_i correctly (malicious partials are caught by PartialDecrypt NIZK and abort the open). The aggregator correctly runs Verify and the chosen robust aggregator.

Claims: ML-KEM-768 transport (FIPS 203 Level 3) [5]; dual-norm soundness for oversized / sparse witnesses under SIS + Unruh/RO; Unruh Enc-consistency so a proof cannot attach to a mismatched **ct**; threshold privacy *and* open correctness against a malicious partial; transcript non-swap across rounds.

TABLE I
THREAT COVERAGE BY LAYER (COMPOSITION IS THE CONTRIBUTION).

Threat	ML-KEM	Dual-norm	Unruh-Enc	PD-NIZK	Median
HNDL / PQ transit	yes	—	—	—	—
Oversized update	—	ℓ_2	—	—	—
Sparse / ℓ_∞ spike	—	ℓ_∞	—	—	help
Proof/ct split	—	bind	yes	—	—
Malicious partial μ_i	—	—	—	abort	—
Curious aggregator	—	—	—	thr.	—
Sign-flip / backdoor	—	no [‡]	—	—	yes

[‡]In-bound updates may still satisfy (τ_2, τ_∞) ; Table IV reports 0% crypto detection for sign-flip under ℓ_2 alone.

TABLE II
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	—	secure agg. stack
Patel [9]	lattice HE	—	HE FL
Lu et al. [10]	hybrid PQ	—	HE FL
Krum [3]	statistical	—	—
zkML [11]	—	inference ZK	—
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

Scope. We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. ZKFL-PQ targets verifiable norm gating, multi-party decrypt, and robust post-filtering; pure ℓ_2 alone does not stop sign-flip or in-bound backdoors (Table I).

III. RELATED WORK

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Multi-Krum remains effective against low-norm adversaries; ZKFL-PQ therefore composes cryptographic accept with median/Krum rather than relying on ℓ_2 alone. We report Hybrid+mean vs. Hybrid+median to make this separation measurable (Table IV).

IV. PROTOCOL DESIGN

A. Round Structure

Each round t : the server broadcasts $w^{(t)}$. Client i runs local SGD on D_i , quantizes to $\tilde{w}_i$, encrypts under BFV public key pk , proves $\|\tilde{w}_i\|_2 \leq \tau$ bound to $\mathbf{ct}_i$ (and Enc-consistency of coins), and sends an ML-KEM-wrapped payload. The server verifies proofs, threshold-opens each accepted ciphertext, and applies coordinate-wise median (default) before the model step (Algorithm 1).

a) Transport.: ML-KEM-768 encapsulates a session key used to AES-CTR-wrap $(\mathbf{ct}_i, \pi_i)$. This protects in-transit confidentiality against classical and future quantum adversaries on the channel; it does not replace HE confidentiality against the aggregator, which follows from RLWE + threshold opening.

b) Verification checklist.: $\text{Verify}(\pi_i, \mathbf{ct}_i, \tau)$ returns 1 iff: (i) Unruh invertible-RO records are consistent; (ii) Fiat–Shamir/Unruh challenge bits match the recomputed digest of first messages and $\mathbf{ct}_i$; (iii) each session satisfies the SIS algebraic relation against the shared $\mathbf{A}$; (iv) response norms respect the rejection bound B ; (v) Enc-consistency accepts for every BFV chunk. Fail-closed: any failed check drops the client for that round.

B. Quantization and Dual-Norm Gate

Quantize clips coordinates to $[-c, c]$, scales by s , and rounds into $\mathbb{Z}_t$ (artifact defaults $c=1, s=10^3$). Before prove+encrypt, honest clients apply a public ℓ_∞ clip $\tilde{w} \leftarrow \text{Clip}_\infty(\Delta, \tau_\infty)$ and then prove $\|\tilde{w}\|_2 \leq \tau_2$ on the *same* vector that is encrypted. Sparse spikes with $\|\cdot\|_2 \leq \tau_2$ but $\|\cdot\|_\infty > \tau_\infty$ are rejected before opening; Enc-consistency prevents proving a clipped vector while encrypting an unclipped one. Adaptive public τ_2 is updated from accepted-norm quantiles and bound into the next round’s associated data together with a rolling transcript hash $H(\text{prev})$.

C. Bound Norm Language

Definition 1 (Bound language). *Let $C = \text{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w}\|\mathbf{r}) \bmod q'$ and $\mathbf{ct} = \text{Enc}_{pk}(\tilde{w}; \rho)$. Then*

$$\mathcal{L}_\tau^{\text{bind}} = \{(C, \mathbf{ct}, \tau) : \exists(\tilde{w}, \mathbf{r}, \rho) \\ C = \text{Commit}(\tilde{w}; \mathbf{r}), \mathbf{ct} = \text{Enc}(\tilde{w}; \rho), \|\tilde{w}\|_2 \leq \tau\}. \quad (1)$$

a) Σ -protocol [6].: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C\|\mathbf{ct}\|T\|\tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z}\|\mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \leq B$. Rejection sampling yields HVZK in the classical ROM. Acceptance uses only algebraic and norm checks—never a client-supplied Boolean.

b) Unruh-oriented NIZK.: Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default $r=128$; latency-sensitive demos often 4–64) with invertible-RO records $(\rho, H(\rho))$. Challenge bits are extracted from $H(\text{UNRUH}\|\tau\|\mathbf{ct}\|\text{first-msgs})$. All sessions share one SIS matrix $\mathbf{A}$ (not r independent keys), which keeps prove/verify interactive at imaging-scale d (one $256 \times (d+128)$ matvec family instead of r separate keys). Combinatorial uniqueness / 2^{-r} lemmas are machine-checked in Lean 4; executable game hops and a SHA3-instantiated EasyCrypt QROM library with bit-level Keccak-f[1600] lane lemmas (θ -column parity, $\rho/\pi/\chi/t$ locality, packing invertibility, 24-round fold) accompany the artifact (`formal/run_formal_ci.py`).

c) Enc-consistency (Unruh-lifted).: Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ -gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk. We lift this gadget under the same Unruh transform as the norm proof (parallel binary sessions + invertible RO records), closing the QROM-uniformity residual between the two ct-bound proofs.

TABLE III
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768	$n=256, k=3, q=3329$	FIPS 203 L3 [5]
BFV degree	$n=512 / 4096$	Lightweight vs. <code>classic128</code>
BFV moduli	$q \approx 2^{40}, t=2^{12}$	NumPy encoder; optional estimator / SEAL
Threshold	$(t, n)=(2, 3)$ partial decrypt	Server $sk=\perp$
Unruh reps	$r=128$ default	Enc Unruh r_{enc} parallel
Post-ZKP agg	median (default)	Multi-Krum / mean overrides
Dual-norm	$\tau_2 + \tau_{\infty}$ clip	Same vector for Prove+Enc
Open audit	PartialDecrypt NIZK	Abort on bad μ_i
Packing	full-vector chunks	All d coordinates encrypted
SIS commit	shared A	One $256 \times (d+128)$ matrix / prover

Enc-consistency first messages are absorbed into the norm proof’s associated data.

D. BFV Aggregation and Threshold Opening

Additive BFV yields $\text{Dec}(\sum_i \text{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. Full-vector packing encrypts all d coordinates as $\lceil d/n \rceil$ independent ciphertexts under the same pk ; the ZKP covers the concatenated quantized vector, so partial packing cannot hide an unprotected suffix. After key generation, secret coefficients are Shamir-shared and the monolithic sk is erased. Opening uses Lagrange-weighted partial decryptions with smudging: party i returns $\mu_i = c_1 \star (\lambda_i \cdot s_i)$ (plus noise). Each partial is accompanied by a PartialDecrypt NIZK proving knowledge of s_{eff} with $\mu_i = c_1 \star s_{\text{eff}}$; a tampered μ_i fails verify and aborts the open—extending threshold privacy to *open correctness*. Modular negacyclic multiplication with parallel per-chunk encryption keeps full-vector runs interactive on CPU for $d \approx 5 \times 10^4$. An optional TenSEAL/SEAL sidecar runs as FusedSealThresholdHE.

a) *Post-ZKP aggregation.*: Let S be the set of clients that pass Verify. Default aggregation is the coordinate-wise median $\bar{\Delta}_j = \text{med}\{\Delta_{i,j} : i \in S\}$. Multi-Krum and plaintext mean are available as overrides. Median is intentionally applied *after* cryptographic accept: it does not replace the ZKP, and the ZKP does not replace median.

Theorem 1 (Composition soundness and threshold privacy). *Assume SIS binding of Commit, RLWE semantic security of BFV, and RO modeling of $H=\text{SHA3-256}$. Then: (i) honest dual-norm witnesses are accepted except for rejection-sampling aborts; (ii) an oversized or ℓ_{∞} -violating witness is rejected except with Unruh/SIS advantage; (iii) Unruh Enc-consistency plus digesting ct prevent attaching a valid proof to a mismatched ciphertext; (iv) the aggregator and any $t-1$ decryptors learn nothing beyond what t partial opens reveal, and a single malformed μ_i is rejected by PartialDecrypt NIZK (open aborts); (v) round transcripts bind across rounds via $H(\text{prev})$. Combinatorial Unruh uniqueness/ 2^{-r} is machine-checked in Lean 4; EasyCrypt*

Algorithm 1 ZKFL-PQ round (default composition)

Require: $w^{(t)}$, τ , KEM (ek_S, dk_S) , HE pk , decryptor shares

- 1: Broadcast $w^{(t)}$
- 2: **for** each client i **do**
- 3: $\tilde{w} \leftarrow \text{Quantize}(\text{LocalSGD}(w^{(t)}, D_i))$
- 4: $\text{ct}_i \leftarrow \text{Enc}_{pk}(\tilde{w}; \rho_i)$; prove Enc-consistency of ρ_i
- 5: $\pi_i \leftarrow \text{Prove}_{\text{Unruh}}((\text{ct}_i, \tau), \tilde{w})$
- 6: send ML-KEM-wrapped (ct_i, π_i)
- 7: **end for**
- 8: Keep i iff cryptographic verify returns 1
- 9: For each accepted i : threshold-open ct_i to recover Δ_i
- 10: $\bar{\Delta} \leftarrow \text{Median}(\{\Delta_i\})$ $\triangleright$ or Multi-Krum; not plaintext mean alone
- 11: $w^{(t+1)} \leftarrow w^{(t)} + \bar{\Delta}$

TABLE IV
MEAN $\pm$ STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0 $\pm$ 1.0	41.0 $\pm$ 13.2	0%	0%
Clip@ τ	100 $\pm$ 0	40.3 $\pm$ 12.7	—	—
Multi-Krum	100 $\pm$ 0	100 $\pm$ 0	—	—
HE-only	22.1 $\pm$ 2.1	19.8 $\pm$ 9.5	0%	0%
ZKP-only	84.9 $\pm$ 30.2	40.9 $\pm$ 13.2	97%	0% [‡]
Hybrid+mean	87.0 $\pm$ 26.0	40.1 $\pm$ 12.6	97%	0% [‡]
Hybrid+median	100 $\pm$ 0	100 $\pm$ 0	97%	0% [‡]

[‡]Sign-flip respects τ — ℓ_2 proofs must accept (0% crypto detection). Hybrid+mean keeps plaintext mean after accept; Hybrid+median is the default composition and restores accuracy [12].

theories instantiate H as SHA3-256 with Keccak lane lemmas (formal/run_formal_ci.py).

b) *Implementation invariants.*: The public repository [12] realizes Algorithm 1 with: (i) identical prove/encrypt dimension d ; (ii) ciphertext bytes in Unruh challenges; (iii) crypto-only accept (no trusted Boolean); (iv) full-vector packing; (v) Enc-consistency; (vi) fused SEAL+threshold HE; (vii) default median; (viii) one shared SIS matrix per Unruh prover. Scripts pin seeds and emit JSON under `results/`.

V. EVALUATION

A. Synthetic Multi-Baseline Study

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet $\alpha=0.5$), $N=5$, MLP with $d=108,996$, 10 rounds, seeds $\{42, \dots, 46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@ τ , Multi-Krum ($f=1$), HE-only, ZKP-only, Hybrid+mean (privacy path without robust agg), and Hybrid+median (default composition). Metrics: final accuracy and cryptographic detection rate (fraction of malicious updates rejected by Verify).

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table IV, Fig. 1–2). Sign-flip within τ is accepted by pure ℓ_2 and

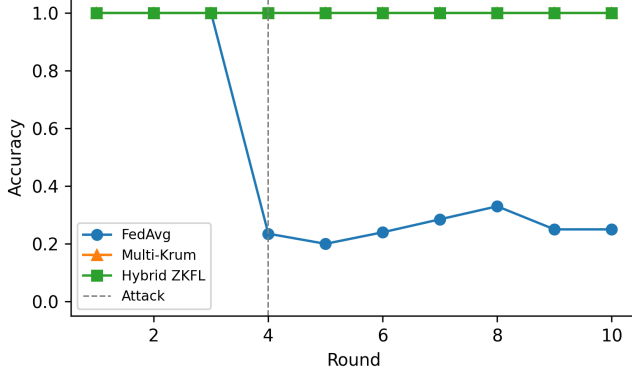


Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

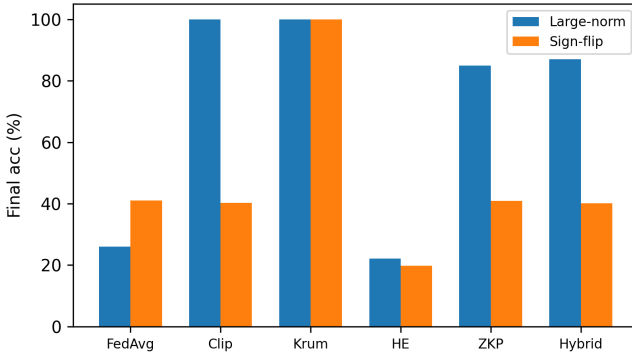


Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

Hybrid+mean (0% crypto detection, Table I), while Multi-Krum and Hybrid+median recover accuracy by pairwise or coordinate-wise robust aggregation after cryptographic accept. The Hybrid+mean vs. Hybrid+median contrast is the empirical punchline for composition: privacy-preserving open without robust aggregation still fails on in-bound flips.

a) Takeaway.: Cryptographic detection is the right metric for oversized attacks; final accuracy after robust aggregation is the right metric for in-bound attacks. Reporting only one of the two understates the threat model in Table I.

B. Scale, Medical, and Backdoor Studies

A compact-model scale study with $N=20$ clients and $T=30$ rounds reproduces the same qualitative pattern under both attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator. Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with $d=1554$), the innovation stack—Unruh ℓ_2 ($r=16$ in the pack demo; library default 128), Unruh Enc-consistency ($r_{\text{enc}}=4$), PartialDecrypt NIZK,

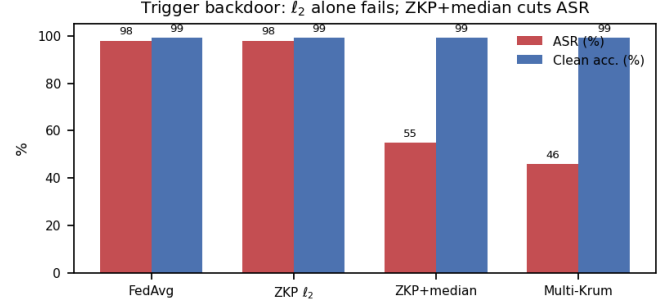


Fig. 3. Backdoor ASR vs. clean accuracy under FedAvg, ℓ_2 -ZKP, ZKP+median, and Multi-Krum.

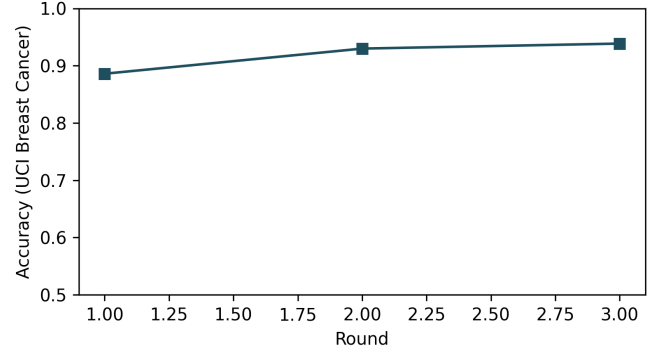


Fig. 4. UCI Breast Cancer accuracy under the full target stack.

adaptive τ_2 , transcript binding, ℓ_∞ clip, full-vector threshold HE, and post-ZKP median—yields accuracies $89.5\% \rightarrow 93.0\% \rightarrow 92.1\%$ while rejecting the oversized client on the attack round; public τ_2 adapts $8 \rightarrow 4.3 \rightarrow 2.3 \rightarrow 2.0$ (results/innovation_pack_results.json). Mean round time is $\approx 20\text{--}23$ s on a laptop CPU for this denser proof surface. A sparse-poison ablation ($\|\cdot\|_2 \leq \tau_2$ but $\|\cdot\|_\infty > \tau_\infty$) shows dual-norm **100%** reject vs. **0%** for ℓ_2 -only, with $\approx 265\times$ lower post-aggregation spike on the poisoned coordinate. Full-resolution PneumoniaMNIST /ConvNet28 results from Sec. V-B remain complementary imaging smoke tests.

Trigger backdoors whose poisoned updates remain inside (τ_2, τ_∞) are still only partially mitigated by median (ASR $\approx 55\%$ vs. $\approx 98\%$ for FedAvg/ ℓ_2 ; Fig. 3); dual-norm removes sparse spikes *before* opening, while median remains necessary for diffuse in-bound poisons.

C. Communication Cost

Full-vector HE dominates bandwidth. Uplink scales as $O(N[d/n]n \log q)$ bits plus Unruh $O(r \cdot (d + \ell))$. Table V reports measured artifact payloads; increasing r grows the proof component linearly, while `classic128` ($n=4096$) roughly doubles ciphertext volume for a fixed d . Operators can therefore trade Unruh soundness bits against uplink and CPU without changing the HE packing layout.

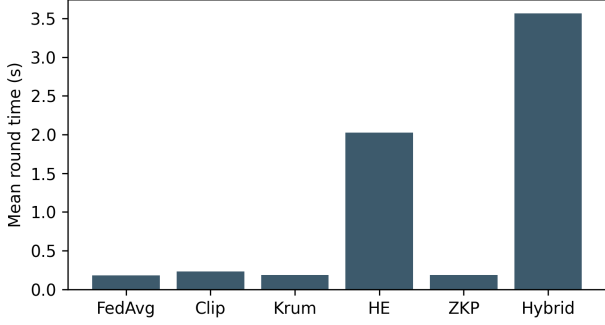


Fig. 5. Mean round time (synthetic hybrid $\sim 20\times$ FedAvg).

TABLE V
PAYLOAD UNDER UNRUH/ n SETTINGS (ARTIFACT ACCOUNTING).

Setting	Msg/round	Notes
Synthetic hybrid	~ 100 KB	Reference accounting
UCI target ($r=64$, $n=512$)	$\approx 4.7\text{--}5.9$ MB	Measured JSON
Medical $r=128$, $n=512$	higher Unruh share	Linear in r
Medical $r=128$, $n=4096$	$\approx 2\times$ ct volume	Classic-128 class
ConvNet28 smoke ($r=4$)	≈ 2.4 MB	$d \approx 51.6k$

VI. SECURITY DISCUSSION

Honest clients with $\|\tilde{w}\|_2 \leq \tau$ produce Unruh sessions that pass algebraic and norm checks except for rejection-sampling aborts (bounded retries in the artifact). If $\|\tilde{w}\|_2 \gg \tau$, sessions with challenge bit 1 fail with overwhelming probability under the Gaussian mask design; soundness therefore depends on the fraction of 1-challenges across r repetitions. The library default $r=128$ targets a ≈ 128 -bit combinatorial Unruh class; demos may reduce r for CPU latency at linear proof-size cost. Binding H to ciphertext bytes, together with the Enc-consistency gadget for ρ , prevents a benign proof from being attached to a different ciphertext. BFV semantic security (RLWE) hides individual updates from parties lacking sk ; with Shamir (t, n) , the aggregator and any $t-1$ decryptors cannot open. We do not claim differential privacy; sites that need DP can compose mechanisms as in Beskar [8].

a) Attack games (informal).: *Oversized/sparse*: adversary wins if $\text{Verify} = 1$ on a witness violating (τ_2, τ_∞) . *Split*: proof for **ct** verifies against **ct'** $\neq$ **ct**. *Malicious partial*: forged μ_i is accepted. *Curious open*: aggregator + $< t$ decryptors recover an honest plaintext. Theorem 1 bounds these under SIS/Unruh/RO, PartialDecrypt NIZK, and RLWE+Shamir.

b) Layering and deployment.: Table I is the scientific punchline: no single primitive meets the joint medical threat model. In a hospital consortium, each site runs a client; a neutral coordinator verifies Unruh norm + Unruh Enc-consistency + dual-norm checks; t -of- n share holders open with PartialDecrypt NIZK (abort on malice); median yields the round update; adaptive τ_2 and the transcript hash are published for the next round. Daily/weekly training absorbs the denser

proof surface (≈ 20 s/round on UCI in the innovation pack). Imaging-scale d remains interactive for Unruh+HE when SIS keys are shared and BFV mul is modular; local SGD still dominates ConvNet28 wall-clock.

VII. LIMITATIONS AND REPRODUCIBILITY

Bit-level Keccak- $f[1600]$ lane lemmas are EasyCrypt-axiomatized with a FIPS executable checker; full discharge without the Python oracle remains future work. Unruh Enc-consistency multiplies Enc prove cost by r_{enc} ; demos use small r_{enc} for CPU budgets while the library default keeps large r for the norm proof. PartialDecrypt NIZK is a research-stack FS proof of the linear relation $\mu = c_1 \star s_{\text{eff}}$ (not MPC-in-the-head). Demo HE uses $n=512$; $n=4096$ is available via `ZKFL_HE_PRESET`. Median+dual-norm reduce but do not eliminate diffuse backdoors—Beskar-style DP remains complementary. Reproduce via `python experiments/run_innovation_pack.py`, `run_target_protocol.py`, and `formal/run_formal_ci.py`.

VIII. CONCLUSION

ZKFL-PQ advances quantum-resilient medical FL by making the ct-bound stack QROM-uniform (Unruh Enc-consistency), decryptor-accountable (PartialDecrypt NIZK), dual-norm gated, transcript-bound, and adaptively thresholded—then composing with threshold BFV and median. Empirically, sparse ℓ_∞ spikes are rejected cryptographically (100% vs. 0% under ℓ_2 -only), oversized clients are still caught, and UCI accuracy remains competitive under the denser proof surface ($89.5\% \rightarrow 93.0\% \rightarrow 92.1\%$). The artifact is complementary to Beskar-class secure aggregation. Artifact: <https://github.com/edlansiaux/pq-zkfl-medical>.

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